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Literature Survey Report

Ref. No.	Year of Publication	Methodology Used	Dataset Used	Performance Metrics	Key Contributions	Limitations
[1]	2026	SeerRAG: Uncertainty Navigator (UN) for adaptive retrieval gating; Modality Discrimination (MD) engine for text/visual/hybrid routing; Cross-Modal Synergistic Generation (CMS) for final response.	LongDocURL; ARC-Easy.	Accuracy: 67.3% (Qwen2-VL-7B), 55.1% (LLaVA-7B); 97% redundant retrieval filtered; latency 7.6s (2.3x speedup).	Dynamic routing triggers retrieval only when model is uncertain; modality-aware routing adapts to query type; balances accuracy and efficiency.	Index is static and uniform; no adaptive indexing at document construction time.
[2]	2026	Dynamic multi-granularity indexing with adaptive semantic chunking (cosine similarity); 5W1H hypothetical QA generation for semantic alignment; BM25 + DPR hybrid retrieval with re-ranking.	HotpotQA; SQuAD.	HotpotQA: EM=37.01, F1=47.53; SQuAD: EM=35.12, F1=45.69; Retrieval Precision=90.1 %.	Adaptive chunking resolves fixed-length fragmentation; hypothetical QA aligns query-knowledge semantics without fine-tuning.	Text-only chunking; ignores visual features, table density, or page layout structure.
[3]	2025	MGA-KGF: Multi-granularity knowledge graph fusion (fine/medium/coarse levels) using TransE-GAT dual-mode fusion and load-aware hot/cold storage indexing.	Smart Grid data (SCADA, fault alarms, load records, O&M reports).	Qualitative improvements in timeliness and cross-granularity reasoning; supports 100K+ access requests/sec.	Validates fine/medium/coarse granularity representation for heterogeneous multi-source data fusion across domains.	Power-engineering domain only; no document retrieval benchmarks; techniques do not transfer directly.
[4]	2026	Multigranularity Fusion (MGF): joint page-level + element-level + answer-guided contrastive losses with granularity-	PCKB (95 construction docs, 22,854 QA pairs).	NDCG@5: 80.42%; Recall@5: 92.73%; element-level gains of 5.5–17.8%.	Joint page and element-level contrastive optimization improves both granularities; answer-guided	Fixed two-level hierarchy applied uniformly; no adaptive depth based on per-page complexity.

		specific projectors for vision-language retrieval.			loss improves domain fidelity.	
[5]	2025	LLM-Centric RAG with hierarchical memory (token/passage/document levels); entropy and variance-based uncertainty estimation for confidence-controlled generation.	CISI dataset.	QA Accuracy: 77.8%; Recall@5: 92.0%; NDCG@5: 90.0%; Factuality: 0.72.	Proves multi-granularity indexing outperforms single-level; entropy/variance filtering suppresses hallucinations in high-stakes domains.	Predefined fixed hierarchy applied uniformly; no adaptive construction based on per-page content.
[6]	2025	AMR-CSI: Indexing Decision Router using layout detection (PubLayNet) and Visual Dominance Score (VDS) to auto-select visual vs. textual indexing pipeline.	2WikiMultihopQA, MultiHopRAG, Vidore (Govt., Health, Energy).	Recall@10: 80.1%; Latency: 253ms; Index size reduced >12x (2.3 GB to 183 MB).	First data-driven framework for automated collection-level modality selection; closes gap to expensive hybrid oracle by <2%.	Binary collection-level decision only; no per-page index depth assignment.
[7]	2023	Heterogeneous GNN with word/sentence/topic nodes (LDA + TF-IDF), GATv2 dynamic attention, and LSTM-based adaptive depth for extractive summarization.	CNN/DailyMail; Multi-News.	CNN/DailyMail: R-1=43.70, R-2=20.19, R-L=39.86; Multi-News: R-1=46.75, R-2=16.47, R-L=42.82.	Topic nodes capture cross-sentence dependencies; adaptive depth/width improves multi-document summarization quality.	Text summarization only; no visual/structural awareness; fixed three-node-type structure.

References

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Signature of the Supervisor **Signature of HOD**